DataGenie

Project Team

Hammad Qaiser	21I-2463
Muhammad Moeed Asif	21I-0483
Muhammad Jamaludin Yahya	21I-0808

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Supervised by

Mr. Shams Farooq

Department of Computer Science

National University of Computer and Emerging Sciences
Islamabad, Pakistan

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Chapter 1

Introduction

DataGenie is an innovative project that aims to change quality assurance in the textile industry by incorporating powerful AI technology. The research focuses on creating synthetic images of fabric defects to train a defect detection system, giving textile producers a more efficient and cost-effective alternative for spotting flaws in textile products.[3]

1.1 Purpose of the Project Proposal

The goal of this project proposal paper is to propose a detailed plan for developing DataGenie, a data augmentation and flaw detection system for the textile sector. This document shows the major aims, methodology and importance of the project in boosting quality assurance processes through the use of AI-driven technology. Moreover, this proposal will present the detailed design of the system both theoretically and diagrammatically.

1.2 Background of the system

In textile industry the defect detection is very important for waste minimization, cost reduction, customer satisfaction and efficient production. Normally, the process of detecting fabric defects such as tears, stains, wefts and holes has been mainly relying on physical inspection, which is time consuming, labor intensive, and there is chance of error or omission. Moreover, gathering big datasets of defective fabrics to train AI-based defect detection systems is difficult since the textile producers may not always have access to a diverse range of defective samples or the producer is a beginner in industry and does not have any source for getting datasets. DataGenie aims to address these issues by presenting a new AI-powered system that creates synthetic images of damaged fabrics, which can then be used to train defect detection algorithms, specifically YOLO. The system will use a Stable Diffusion model to generate very realistic synthetic representations of fabric defects depending on user input. These images will be used for training data for deep learning algorithms, allowing them to spot flaws more precisely. The system will also include a real-time flaw detection method based on the YOLO model, which will be implemented on a user-friendly web application. Users can input fabric images, and the model detects and highlights flaws in real time, allowing manufacturers to enhance their quality control operations.[1]

1.3 Problem Statement

Traditional QA methods rely heavily on real world data, which can be difficult to obtain, expensive to acquire, and often contains sensitive information that cannot be shared freely. Moreover, real data sets may not always represent the full range of potential defects that need to be tested, leading to inaccurate results and overlooked defects. The lack of sufficient and representative data can severely hinder the effectiveness of quality assurance processes, leading to delays, increased costs, and compromised product quality.[2]

1.4 Scope

The DataGenie project aims to create a reliable system for detecting fabric defects and producing synthetic data for the textile industry. This solution is intended for AI engineers working in quality assurance. It includes an application that uses a Stable Diffusion model to generate realistic synthetic images of fabric defects. To address the issue of limited datasets for training defect detection models, the generated synthetic data will train a YOLO-based model for real-time detection of fabric defects. A verification model will also be developed to compare synthetic defect datasets to real-world data in specific defect categories. This model ensures the accuracy of synthetic data by demonstrating that any difficulty distinguishing between real and synthetic defects reflects the generated data's precision. Furthermore, the system will accept natural language prompts, allowing users to specify defect types and orientations. These inputs will be processed by a large language model (LLM) trained on textile specific data which will result in synthetic datasets. To streamline the quality assurance process, all functionalities will be integrated into a user- friendly web application. The system will also include predictive analytics, which will use prediction analysis algorithm to predict fabric defects based on past data or production conditions like temperature and humidity. This analysis capability will aid in defect prevention by predicting when and where they are most likely to occur, providing an additional layer of quality assurance to the workflow.

1.5 Modules

1.5.1 Module 1: Dataset Annotation and Stable Diffusion Model Development

In this module, the focus is on manually annotating and connecting fabric defect datasets. The annotated data will be used to train the Stable Diffusion model, which generates synthetic data. A basic web app user interface will also be developed.

- **Manual Data Annotation:** Annotate and label fabric defect images for model training.
- **Stable Diffusion Model:** Train a Stable Diffusion model to generate synthetic fabric defect images.

1.5.2 Module 2: NLP Model Training and Database Setup

This module involves training an NLP model for processing text-based user inputs and organizing the annotated and synthetic datasets into a database.

- **NLP Model Training:** Train the model to interpret and process user input for fabric defect generation.
- **Database Setup:** Store the dataset efficiently for retrieval and processing.

1.5.3 Module 3: YOLO Model Integration and Web App Development

The primary objective of this module is to implement the YOLO model for detecting fabric defects and optimizing it for accuracy. Additional features of the web app will also be developed in this phase.

- **YOLO Model Implementation:** Integrate and train the YOLO model to detect fabric defects.
- **Web App Optimization:** Enhance the web app for better user interaction and performance.

1.5.4 Module 4: Verification Model and Predictive Analytics

This module introduces a verification model to cross-check the accuracy of the synthetic data generated. It also focuses on building predictive analytics for fabric defect trends.

- **Verification Model:** Develop a model to validate the GANs and YOLO model's results against real and synthetic data.
- **Predictive Analytics:** Implement analytics for predicting future fabric defects based on historical data.

1.6 User Classes and Characteristics

User class	Description
AI Developer	As the AI Developer, the candidate needs to build and deploy the machine learning that supports the DataGenie project. The primary objectives of the AI Developer will be the Stable Diffusion model for the generation of synthetic data and the YOLO model for defects. It will also involve applying techniques to LLM's with respect to the interpretation of user prompts in order to generate data to suit textile industry requirements. Again, the AI Developer will be coordinating with other team members so that these models can fit well into the overall framework of the application; in addition, the AI Developer will frequently be responsible for performance tuning on these models as well as redesigning them given fresh data or feedback.
Quality Assurance Manager	It's the responsibility of the Quality Assurance Manager to confirm that the synthetic datasets and the defect detection models are operational as expected and up to the benchmark standard. This person will be responsible for evaluating the Quality of synthetic images, which is produced by Stable Diffusion model and the results of the YOLO model that is used for detecting defects. It will be their duty to create test cases, assess the model's fitness for use and ensuring quality of the model at different stages of model creation. The Quality Assurance Manager will offer feedback to the AI Developers to improve model performance and hence will be of great significance. Furthermore, this work entails consultation and collaboration with stakeholders in order to satisfy the quality assurance requirements of the textile manufacturing industries as to the application.
Textile Manufacturer	The Textile manufacturer is the main user of the DataGenie application with purpose to introduce improvement to the quality control of their production lines. A power user class embraces such a user who is in charge of fabric production and quality of the end products. Through this web application manufacturer view the results of the defect detection. Their contribution will be important in the fine-tuning of the application since they will identify which defects are prevalent in their manufacturing processes. This user class will seek to use the aspects of the application that enables a prediction of possible quality problems and which can be prevented.

Chapter 2

Project Requirements

2.1 Event Response Table

EVENT	SYSTEM STATE	RESPONSE
User adds image to the upload tab and clicks upload image	Image is saved in database and fed to Stable Diffusion model	"Image uploaded" screen displayed
User enters prompt in text box and clicks generate button	The input is processed, and label and amount of images are extracted	User navigated to a loading screen
User clicks "show images" button	Generated images are loaded to the screen	Generated images displayed on a screen
User clicks on an image	Image enlarged to full screen with annotation display	Image displayed on full screen
User clicks "delete image" button	Screen displays warning message	Image deleted from database
User clicks "download dataset"	Screen displays size and information of dataset	Images and annotations downloaded locally to device
User clicks "train model" button	Images and annotations fed to YOLO model, and training starts	Navigated to training screen
User clicks "download model" button	System exports PyTorch model	PyTorch model is saved to device

Table 2.1: System Events and Responses

2.2 Functional Requirements

1. **Synthetic Data Generation**
   - The system must be able to generate synthetic fabric defect datasets using a Stable Diffusion model.
   - Users must be able to provide input through prompts to generate specific types of fabric defects.

2. Defect Detection
   - The system must integrate defect detection model to perform real-time fabric defect detection according to the given images.
   - The system must allow users to upload fabric images and receive detection results having the detected defects.

3**. Prompt-Based Defect Generation**
   - Users must be able to specify defect types and orientations through natural language prompts.

4. Model Verification
   - The system must include a verification model to compare the generated synthetic datasets against real world data to ensure the accuracy.
   - The system must validate synthetic data accuracy by determining if the synthetic images resemble to actual defects.

5. **Predictive Analytics**
   - The system must provide predictive analytics based on historical fabric defect data or production conditions like humidity and temperature.
   - Predictive analysis should help identify when and where defects are most likely to occur to assist QA in precautionary measures.

6. **Database Management**
   - The system must store annotated and synthetic datasets in a database for the easy retrieval and use.
   - The system must support the storage of real-time detection results and user given inputs.

7. **Web Application Interface**
   - The system must offer a user-friendly web interface for interacting with all functionalities, which are: uploading images, entering prompts, and viewing defect detection results.

8. **Support for Multiple Defect Types**
   - The system must allow generating and detecting a variety of fabric defects to cover comprehensive quality assurance needs.

9. **Training and Optimization**

- The system must allow for the training and optimization of AI models (Stable Diffusion, YOLO) to improve accuracy and efficiency over time.
- The system should enable model retraining based on new data or feedback.

10. **Data Collection and Pre-processing**

- The system must support data collection and pre-processing functionalities to properly annotate and label fabric defect images for model training.

2.3 Non-Functional Requirements

2.3.1 Reliability

- **NFR-1**: In order to carefully avoid time losses while identifying defects, as well as enriching data, the entire system it is required that the mean time between failures is at least 500 hours..

- **NFR-2**: Any subsequent failures and errors that are executed by the system should be managed in such a way that they spring back up, without disturbing the flow of process which should preferably not take more than 5 minutes.

- **NFR-3** The system should have a capability to return to the previous successful configuration in the case of an error in synthesizing the data or during the identification of defects.
- **NFR-4**:The system should have a backup mechanism which guarantee all the datasets are quite often backed up in the cloud so that they do not get lost.

2.3.2 Usability

- **NFR-5:** The web application will have an easy to use interface which helps the user to first time users to complete the synthetic datasets generation within 10 minutes of interaction with the application.
- **NFR-6:** Feedback to given prompts shall be within the next 2 seconds for the user acknowledgement including submitting prompts, beginning defect detection, or loading datasets as a couple of examples**.**
- **NFR-7:** The system should also have recognizable error messages, and a supplied recovery concerning errors to allow users to handle them within 3 interactions.

2.3.3 Performance

- **NFR-8:** The Stable Diffusion model used in the system shall produce synthetic fabric defect data set within 5 minutes for up to several of these images.
- **NFR-9:** System should be to perform defect detection in the time frame of 5ms per image on a high performance GPU.
- NFR-10:The Large Language Model (LLM) shall be able to reply to the user prompt in the following manners:
  Number of images suggested per defect type within 5 seconds of submission.
- **NFR-11:** The system should be able to support up to 500 concurrent users that does not decrease efficiency.

2.3.4 Security

- **NFR-13:** System should perform all data transmissions between the client and server must be encrypted using TLS 1.2 or higher.

2.3.5 Project Requirements

- **NFR-14:** User sessions should be authenticated using JWT (JSON Web Tokens) and support OAuth 2.0 for third-party integrations.

- **NFR-15:** The system shall enforce strong password policies by requiring at least 8 characters, one uppercase letter, one lowercase letter, one number, and one special character.

- **NFR-16:** The system should automatically log out users after 15 minutes of inactivity to protect against unauthorized access**.**

- **NFR-17:** The system should ensure that only authorized user can access the application i.e the user with authorized account and correct credentials.

2.4 Domain Model

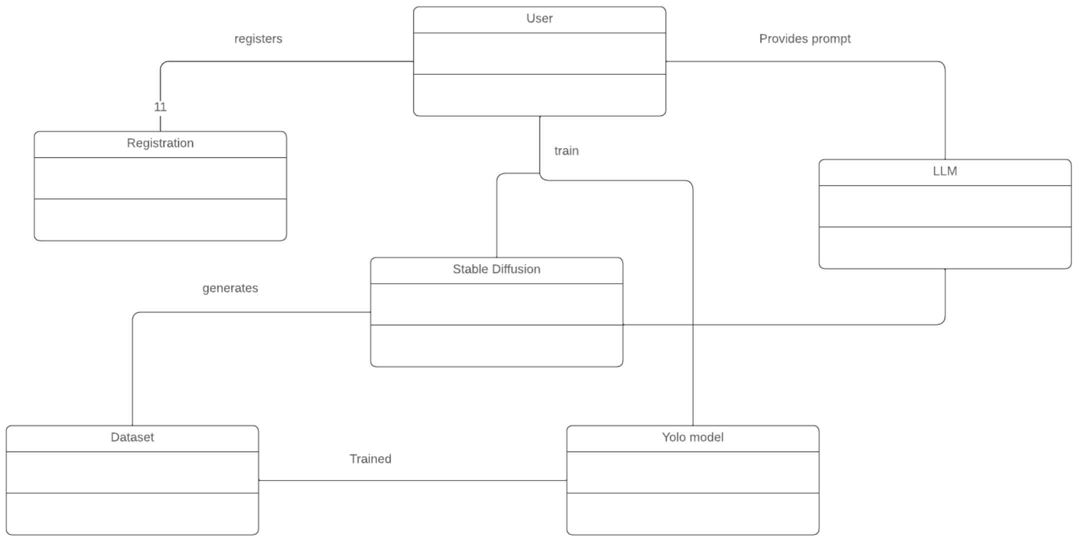


Figure 2.1: Domain model diagram.

Chapter 3

System Overview

Give a general description of the functionality, context, and design of your project. Provide any background information if necessary.

3.1 Architectural Design

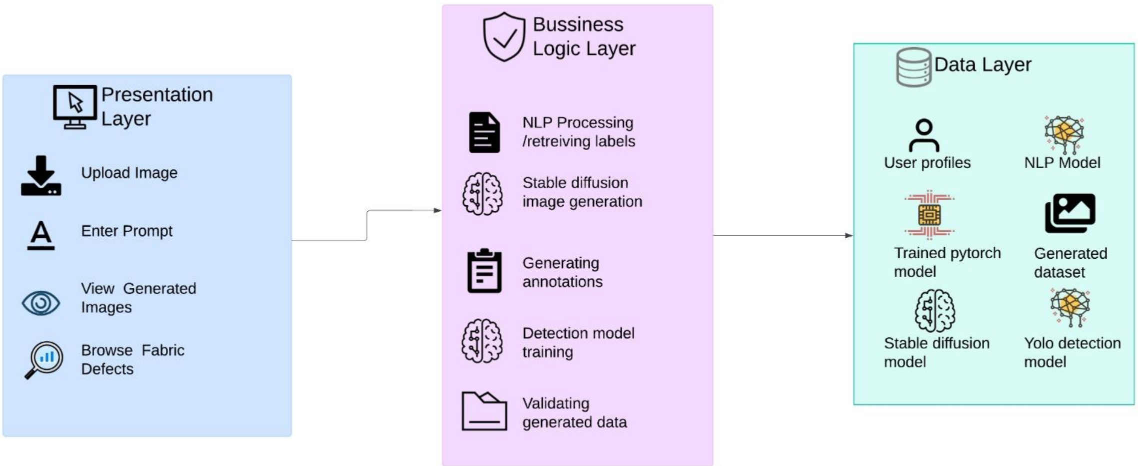


Figure 3.1: Architectural diagram.

3.2 Design Models

3.2.1 Data Flow Diagram

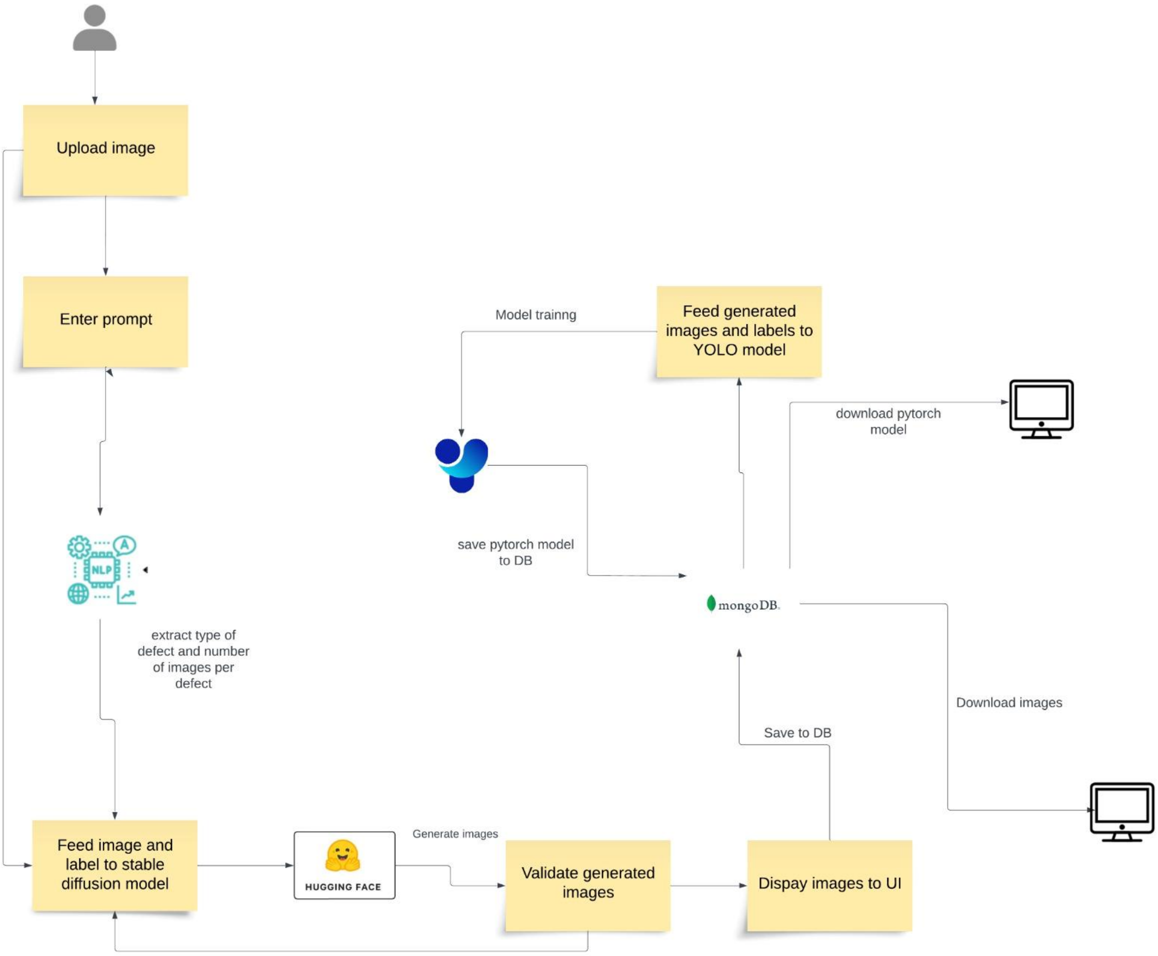


Figure 3.2: Data Flow Diagram

3.2.2 System Sequence Diagram

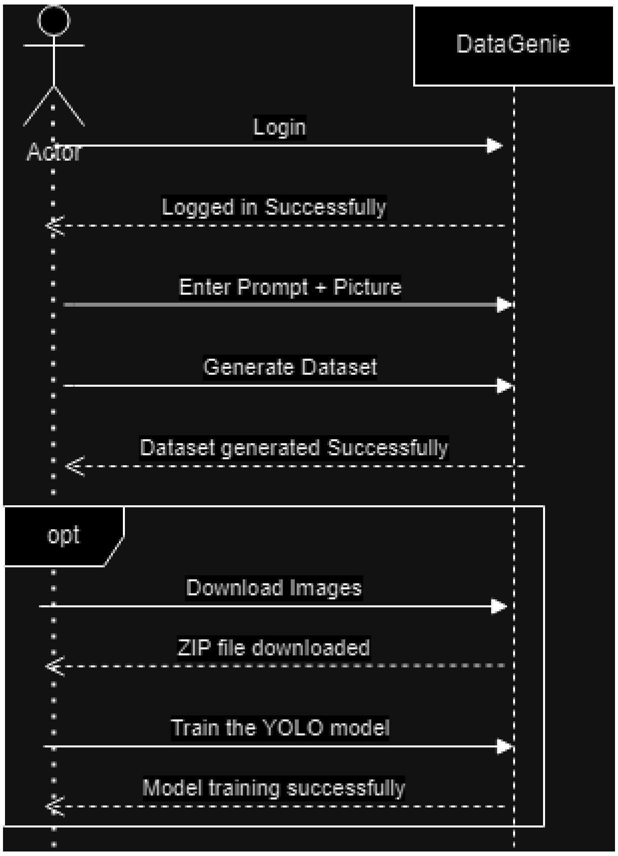


Figure 3.3: System Sequence Diagram.

3.2.3 Activity Diagram

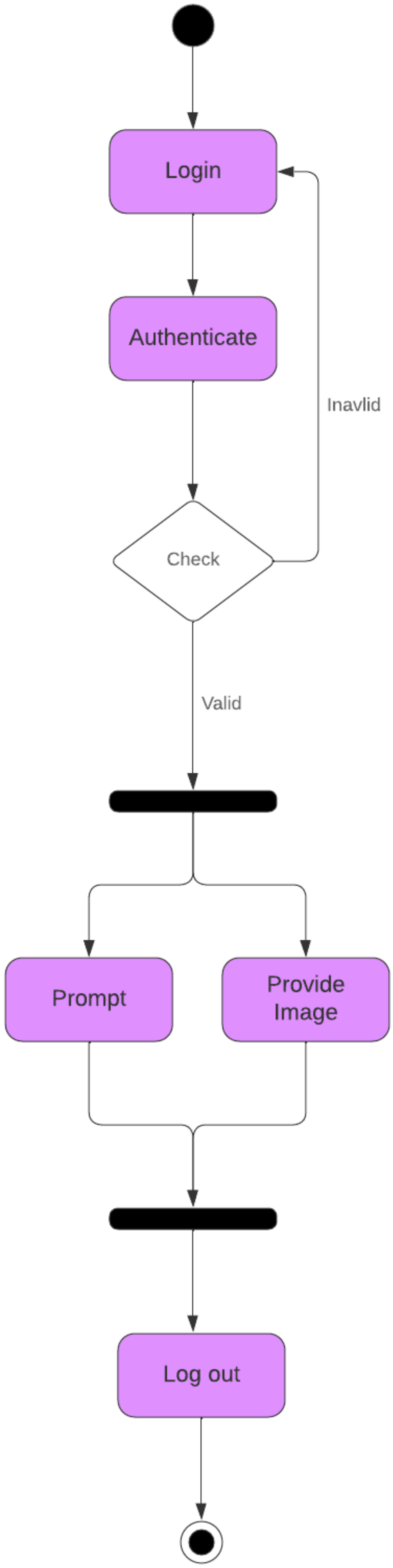
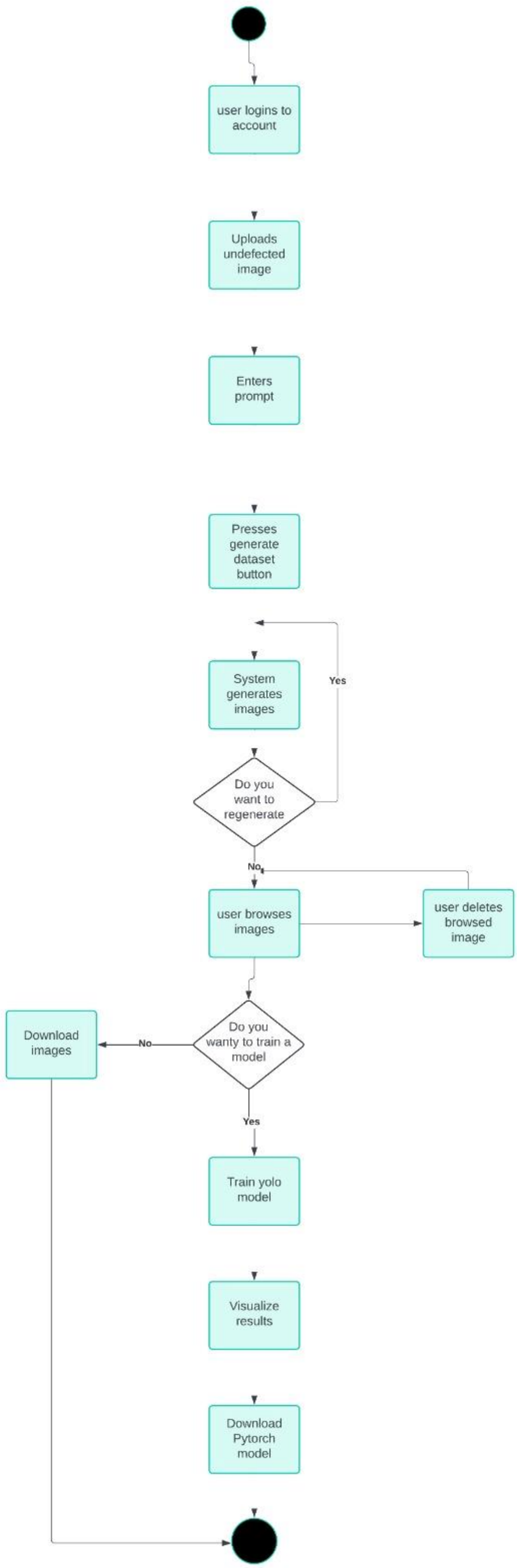

Login
Authenticate
Inavlid
Check
Valid
Prompt
Provide Image
Log out

Figure 3.4: Activity Diagram.

3.2.4 State Transition Diagram


user logins to account
Uploads undefected image
Enters prompt
Presses generate dataset button
System generates images
Yes
Do you want to regenerate
No
user browses images
user deletes browsed image
Download images
No
Do you wanty to train a model
Yes
Train yolo model
Visualize results
Download Pytorch model

3.3 Data Design

The DataGenie project includes multiple entities, which are required for the realization of fabric defect detection and synthetic data generation. The information domain is translated into several structures, through which the information data can be most effectively stored and processed. The storage, process and organization of the major data entities in the framework Is briefly described below.

3.3.1 User

Attributes:

- **name:** The name of the user.
- **id:** An identification number for the user.
- **email:** The user's email address.
- **pass:** The password the user uses for the purpose of identification.

Storage: Holding users' information, the **User** entity resides in the database in a table called *User* and performs authentication with a role-based access control mechanism at the application level.

3.3.2 Dataset

Attributes:

- **image_id:** This is a tag that every image in the dataset should have.
- **annotation:** Information on any changes that may have been made to the images, specifically, annotations.

Storage: The **Dataset** entity is contained within a *Dataset* table, with rows correlating to specific images and their annotations. The structure allows easy location and manipulation of image data.

3.3.3 Notification

Attributes:

- **id:** A unique media identifier assigned to the notification.
- **notification_data:** The information to be delivered in the notification.
- **user_id:** The identification of the **User** who receives the notification.

Storage: Notification updates are logged in a *Notification* table to provide communication and updates to the users regarding their activities or system events.

3.3.4 Prompt

Attributes:

- **id:** A unique identifier for the prompt.
- **prompt_text:** Text input that the user provides as the prompt for the program to analyze.
- **user_id:** A unique identifier for the user creating the prompt.

Storage: Prompts are stored in a *Prompt* table where users input data needed to fulfill their synthetic data generation requests.

3.3.5 Defect

Attributes:

- **id:** A unique identification number assigned to the defect as it is reported to the management.
- **type:** The specific type of defect.

Storage: A *Defect* table is used to store defects and categorize different types of defects that might occur in the fabric.

3.3.6 Fabric

Attributes:

- **id:** A unique number that identifies the fabric.
- **details:** Additional information about the fabric.
- **type:** The category of the fabric.

Storage: Fabric details are recorded in a *Fabric* table, allowing the system to differentiate between different types of fabrics and their respective details.

3.3.7 Data Storage and Organization

The entities are stored in Nosql collections, ensuring that any data is well-organized and easily retrieved. Many of these entities interface with each other; for example, **Users** create **Prompts**, and **Defects** are linked to **Datasets**. These relationships are managed through foreign keys. This structured approach enables efficient querying and manipulation of the data to support the general functioning of the DataGenie application.

3.3.8 Databases and Data Storage Items

Database Management System: The project utilizes a MongoDB database management system, to manage the data.

Image Storage: The images associated with the **Dataset** entity can either be stored in cloud storage services such as S3 or Blob storage or in the database itself. The decision regarding where to store images depends on the volume of data and the frequency of access required.

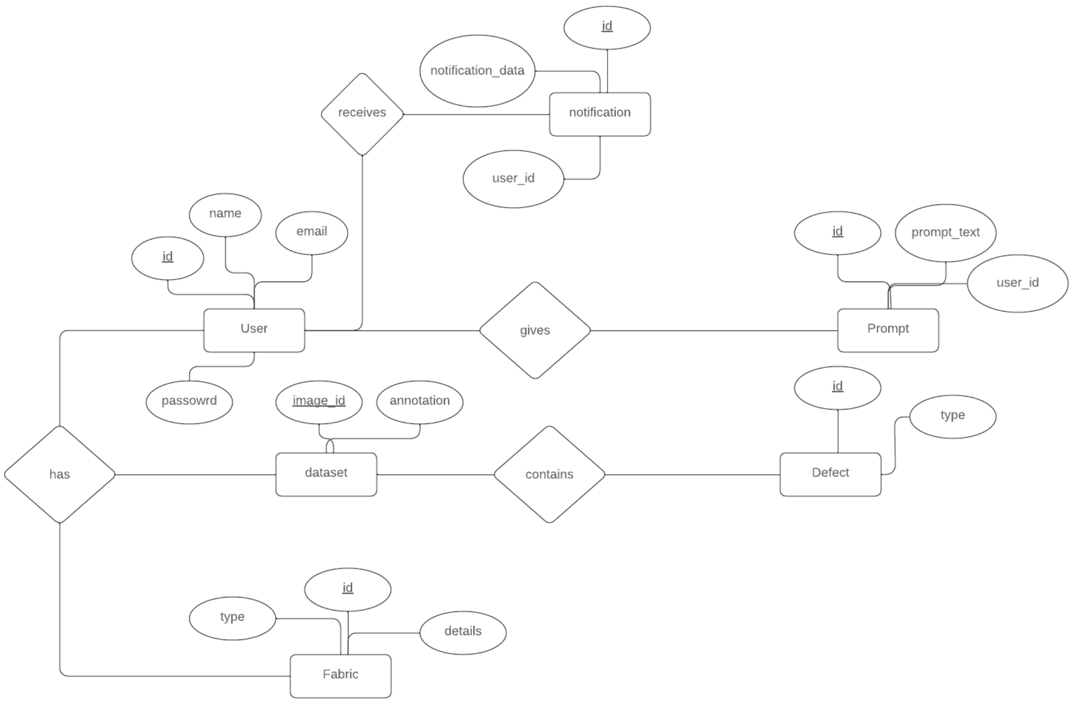


Figure 3.5: Entity Relationship Diagram

Bibliography

[1] Amazon. What is synthetic data. *Synthetic data generation*, 2024.

[2] Microsoft. Synthetic data generation. *Synthetic data generation*, 2024.

[3] Turing. Synthetic data generation: Tools and techniques. *Synthetic data generation*, 2024.